

Enhancing Turkish Reasoning in LLMs

Reward $\Rightarrow$ *judge* $\Rightarrow$ *Enhance*

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MEDPRAI

2026 - July



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Two Systems.

Based on fast memory retrieval and basic iterations. Simple arithmetic operations and well-known phenomena can be an example.

No thinking or pondering involved, direct one-shot answer is given. Most straightforward system.

Before the “thinking” paradigm started in LLMs this was the approach. No “step-by-step” prompting, only direct answers with few-shots.

System I

Time consuming and involves deep pondering. Usually used when confronting a novel or complex problems.

Although time and energy consuming, it is more dynamic and solid approach for problem solving.

Once we started training the models with <think/> tags and “Step by step” prompts, the models entered *System II*.

System II

Who cares?

It is cool that we can imagine LLMs unlocking System II, but why bother?

01

Robustness

LLMs will face novel problems every single day. Activating mechanism that is capable of creative problem solving when needed is a must in fast changing world.

02

Resources

Assuming we can retrain the model with no energy/time constraints, some languages doesn't even have the data to train on.

03

Generalization

We train a model to gain a robust foundation in a specific domain and expect it to generalize from there, without System II it simply can't.

Unlocking System II

- To enhance the model's mathematical foundation SFT was applied.
- To make the model better at deducing its paths DPO is tried.

Train

- Search framework has been conducted :
Reward - Judge - Enhance
- Different combinations of the proposed framework is tested.

Inference

Classical Method

Let's strengthen the model's
Math foundations.

	ARC	GSM8K	Hellaswag	TruthfulQA	Winogrande	Average
Mix-50K	0.48	0.55	0.50	0.53	0.57	0.52
Mix-40K	0.48	0.55	0.50	0.53	0.57	0.52
Mix-30K	0.49	0.57	0.51	0.54	0.56	0.53
Mix-20K	0.47	0.56	0.51	0.54	0.56	0.52
Metamath-50K	0.47	0.55	0.51	0.52	0.57	0.52
Numina-50K	0.47	0.54	0.50	0.52	0.57	0.52
TigerLab-MathInst-50K	0.48	0.47	0.51	0.53	0.56	0.51
Metamath-300K	-	0.49	-	-	-	-
<i>Baseline (Cosmos-DPO)</i>	<i>0.51</i>	<i>0.57</i>	<i>0.52</i>	<i>0.57</i>	<i>0.57</i>	0.58

- Different datasets with different mixes.
- LoRA to preserve model's general abilities.
- No noticeable performance increment.

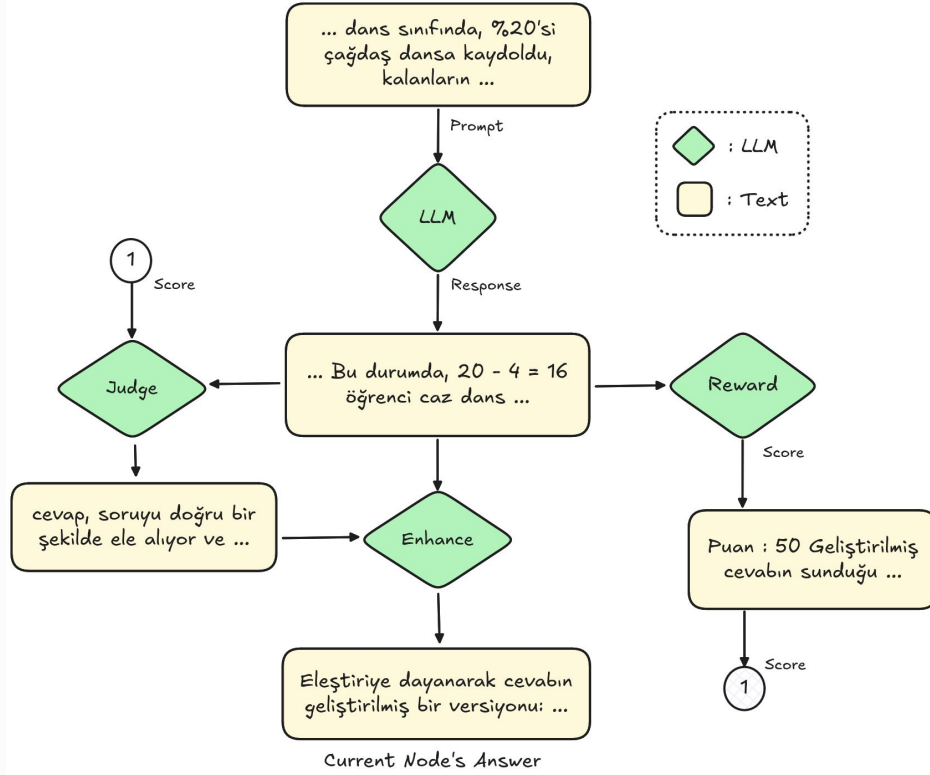
Non-Classical Method

Merged Models	Method	GSM8K
DPO, Metamath-50K	Linear	0.62
DPO, Metamath-50K	Ties	0.57
DPO, Numina-50K	Ties	0.58
DPO, Metamath-50K	Breadcrumbs	0.54
DPO, Metamath-50K	Slerp	0.25
Metamath-50K, DPO	Slerp	0.26
DPO, Metamath, Numina	Linear - Same Time	0.60
DPO, Metamath, Numina	Linear - Binary	0.61
DPO, Metamath-50K	Evolve - Linear (24 Steps)	0.62
DPO, Metamath-300K	Linear	0.63
<i>Baseline (Cosmos DPO)</i>	-	0.57

Let's strengthen the model's Math foundations.

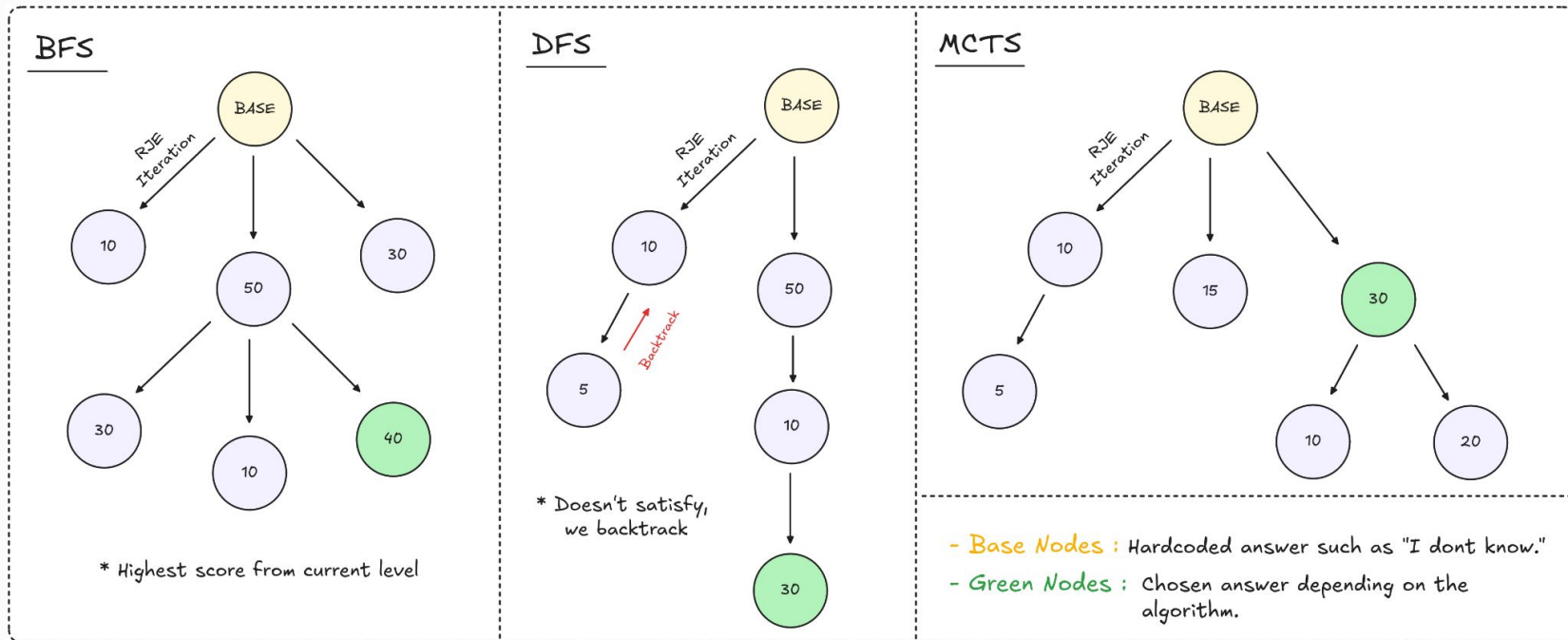
- Using simple Linear merge algorithm the model's mathematical abilities increased.
- The eval is done using the popular framework lm-eval-harness.
- Now, we can try search methods.

System II For LLMs



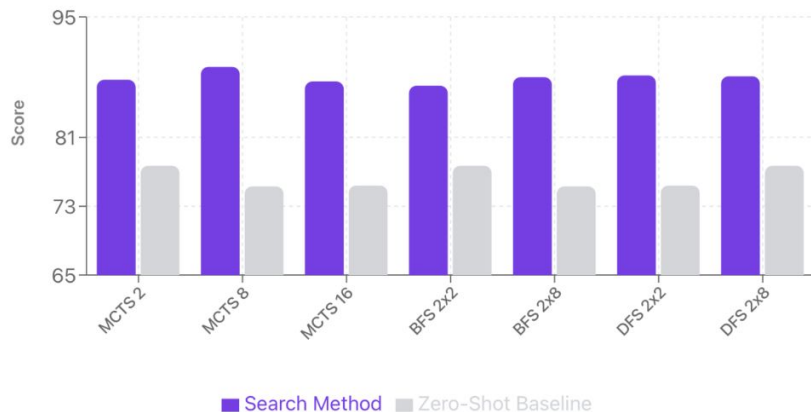
- Simple mathematical foundation is needed to understand instructions.
- This results in a single node in the Tree of Thoughts.
- Ability to reflection on the answer is needed.
- Can be done in different ways.

System II For LLMs



System II For LLMs

Qwen2.5-7B Model



Average improvement: **+11.6 points**

Cosmos-DPO Model



Average improvement: **+13.8 points**

Dissecting System II

- Our framework consists of three modules : Reward, Judge & Enhance.
- But which was the most dominant in this phase ?
- We can use a “smarter” model to supervise the smaller one, which means offloading one of the modules to the smarter one

Why we obtained such results ?

- Since directly “enhancing” the solution by a better model inherits the fallacy that the main model won't solve we it we discard it.

Client	Reward	Judge	Enhance
Client-0	Self-Reward	Self-Judge	Self-Enhance
Client-1	GPT	GPT	Self-Enhance
Client-2	Self-Reward	GPT	Self-Enhance
Client-3	GPT	Self-Judge	Self-Enhance

Dissecting System II

Why we obtained such results ?

	Cosmos-DPO			Qwen2.5-7B		
Client 0	75	92	89	77	76	75
Client 1	92	88	89	85	82	85
Client 2	85	87	86	79	83	90
Client 3	85	69	78	80	76	83
	BFS 2x2	BFS 2x8	BFS 2x16	DFS 2x2	DFS 2x8	DFS 2x16

- The deeper we go in searching the less “supervision” is needed.
- Model’s who doesn’t score their own answers fail to solve the problem without a criticism from the same resource.

Boosting System II

- Models with better ability to reflect on their thoughts perform better in this RJE pipeline.
- Forcing models to think in this systematic manner can be done through DPO training.
- Choosing the superior model as Gemma2-27B and given 12K math questions we can run the pipeline to obtain :
 - 12K Reward.
 - 12K Judges
 - 12K Enhancements
- The same is done for the regular model to form a dataset used in DPO training.
- Unfortunately this approach didn't result in success.

Check it out.

Project Website.

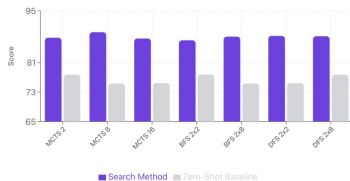
Enhancing Reasoning in Turkish LLMs with Search-Based Inference

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MEDPRAI 2025

[Paper](#) [Code](#) [Dataset & Models](#) [arXiv](#)

Qwen2.5-7B Model



Average improvement: +11.6 points

Cosmos-DPO Model



Average improvement: +13.8 points

- For code, datasets and models visit this website.

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Any questions? Ask away!